

Instructions to the Students :

1. Each question carries 12 marks.
2. Question No. 1 is compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per CBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculator is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		CO	Marks
Q. 1	Objective Type Questions. (Compulsory Question with one marks each)	[2,2]	(12)
(i)	Transfer function is defined as (a) ratio of input to output, (b) ratio of Laplace transform of input to Laplace transform of output, (c) ratio of output to input, (d) ratio of Laplace transform of output to Laplace transform of input		
(ii)	The error detector element in control system gives (a) sum of reference and feedback signal, (b) sum of reference and error signal, (c) difference of reference and feedback signal, (d) difference of reference and feedback signal		
(iii)	The generator constant K_g for a generator given at 1000 rpm is 50 V/A. If the generator is made to run at 1500 rpm, the new constant will be (a) 33.3 V/A, (b) 33.5 V/A, (c) 150 V/A, (d) 100 V/A		
(iv)	The expression $c(t) = e^{-0.6t} \sin 0.8t$ is impulse response of a system, the damping ratio and natural frequency of oscillations will be (a) 0.6, 0.6 rad/sec, (b) 0.6, 1 rad/sec, (c) 0.8, 0.6 rad/sec, (d) 1, 0.8 rad/sec		
(v)	For the system $\frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}$ the damping factor ζ and damped frequency of oscillations ω_d will be (a) $\zeta=0.6$, $\omega_d=5$, (b) $\zeta=0.4$, $\omega_d=6$, (c) $\zeta=0.5$, $\omega_d=3$, (d) $\zeta=0.3$, $\omega_d=5$		
(vi)	The closed-loop poles of open-loop transfer function $G(s)H(s) = \frac{K}{s(s+2)}$ at $K=2$ are (a) $1 \pm j1$, (b) $-1 \pm j1$, (c) $-2 \pm j1$, (d) $2 \pm j1$		
(vii)	For 0% overshoot, the damping ratio for a second order system is (a) <1 , (b) $=1$, (c) >1 , (d) zero		
(viii)	The type of system of transfer function $G(s)H(s) = \frac{K}{s^3 + 2s^2 + 3s}$ is (a) 1, (b) 1, (c) 3, (d) zero		
(ix)	An increase in damping ratio (a) increases rise time, (b) decreases rise time, (c) does not affect rise time, (d) keeps the rise time within limits		
(x)	The Routh-Hurwitz stability criterion gives (a) relative stability, (b) absolute stability, (c) gain margin, (d) phase margin		
(xi)	A second-order control system with $\zeta < 0$ is always (a) stable, (b) unstable, (c) marginally stable, (d) inherently stable		
(xii)	The location of closed-loop conjugate pair of poles on $j\omega$ axis indicates that the system is – (a) stable, (b) marginally stable, (c) unstable, (d) critically stable		

Q. 2 Attempt the following.

(12)

- (a) Compare open-loop system with feed-back control system. [2,2]
- (b) Draw a generalized block diagram of feedback control system and briefly describe its various components. [2,2]

Q. 3 Attempt the following.

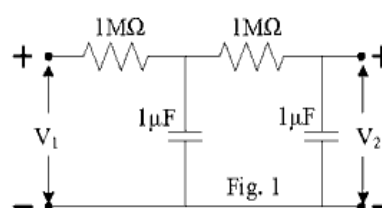
(12)

- (a) Write and briefly describe Mason's gain formula. [3,3]
- (b) Obtain the transfer function of armature-controlled DC servo motor with clearly defining the terms in the equation and assumptions made. [3,3]

Q. 4 Attempt any two of the following.

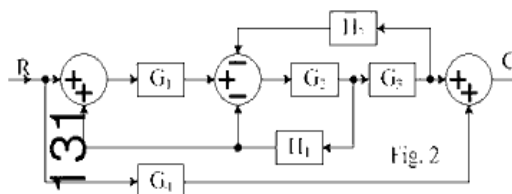
(12)

- (a) Obtain transfer function for the electrical network shown in fig. 1



[3,2]

- (b) Obtain transfer function for the system shown in fig. 2 using block diagram reduction technique.



[3,2]

- (c) Derive the expression for $c(t)$ for a standard second-order unity-feedback system when unit-step input is applied to that system. [3,3]

Q. 5 Attempt any two of the following.

(12)

- (a) Applying Routh-Hurwitz criterion, comment on the stability of the system having characteristics equation as $S^6 + S^5 + 5S^4 + 3S^3 + 2S^2 - 4S - 8 = 0$ [4,4]
- (b) Briefly describe the steps to construct Bode-Diagram (magnitude and phase plot) for any given system. [2,4]
- (c) Describe the Nyquist stability criterion with necessary example. [2,4]

Q. 6 Attempt any two of the following.

(12)

- (a) For a certain control system, the transfer function given by $\frac{Y(s)}{U(s)} = \frac{1}{s^2 + 2s + 5}$, obtain the state-space representation for the system. [3,4]
- (b) Explain Derivative (D) controller with necessary diagrams, graphs, and equations. [2,3]
- (c) Briefly describe the term Controllability with necessary equation / example. [2,4]